

HONGZHE XIE

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EDUCATION

University of California, Davis

Expected May 2028

Davis, CA

University of California, Merced

Computer Science and Engineering | Major GPA 3.992/4.0

Merced, CA

EXPERIENCE

Full-Stack Software Engineer

Oct 2025–Present

School of Engineering, University of California, Merced

Merced, CA

- Architected and now operate the program's official production platform (Django REST Framework, React 19 + TypeScript + Vite), which runs partner intake, team matching, and milestone tracking for capstone collaborations between industry sponsors and student teams.
- Built the platform's block-based CMS, registration flow, barcode ticketing, and check-in, giving staff an auditable trail of every administrative action.
- Designed a dual-tier agentic workflow separating authenticated admin operations from read-only public event queries, isolated by RBAC, bounded context, per-IP token budgets, and privilege-escalation guards.
- Built an AI-assisted phishing detector validating SPF/DKIM/DMARC and sender alignment plus punycode, typosquat, and URL heuristics, with Amazon Bedrock review over a deterministic rules-only fallback.
- Mentor undergraduate teams on scoping, system design, and code review, holding student work to production standards for industry sponsors.

Computer Vision / AI Engineering Intern

Jul 2023–Oct 2023

China Academy of Building Research

Shanghai, China

- Built a Python semantic-segmentation pipeline that identifies deployable rooftop photovoltaic area in aerial imagery of rural villages.
- Annotated and curated the training dataset, then converted segmented rooftop area into estimated annual solar generation capacity.

SELECTED PROJECTS

TokenRouter

2026

Go, React, OpenAI/Anthropic/Google APIs, GORM, Redis, Casbin, Docker

- Built an OpenAI-compatible LLM gateway that converts between provider protocols, routes by priority and weight with failure-aware retries, and reserves and settles token quotas concurrency-safely.
- Hardened relay and control planes with SSRF-safe outbound dialing, JWT refresh rotation with replay detection, Casbin RBAC, and WebAuthn/TOTP step-up verification; validated quota accounting under race-condition tests.

MobileID

2025–2026

Python, Django, Vue, PostgreSQL, DynamoDB, Docker, GitHub Actions

- Built a Django/Vue identity platform for PDF417 barcode generation, profile management, and device-aware sessions on PostgreSQL and DynamoDB.
- Layered authentication defenses: HTTP-only cookie JWTs, CSRF validation, refresh-token rotation and blacklisting, immediate session revocation, and Argon2 hashing.
- Gated CI on Bandit, Safety, Trivy, and dependency audits at a 70% coverage floor, plus Playwright E2E suites and Docker/Vite build validation.

TECHNICAL SKILLS

Languages: Python, Go, C/C++, TypeScript/JavaScript, SQL

Web & Data: Django REST Framework, React, Vue, Vite, PostgreSQL, MySQL, DynamoDB, Redis

AI & Agents: Google ADK, Gemini, Amazon Bedrock, agent orchestration, structured tool calling, LLM application security

Security: WebAuthn/FIDO2, TOTP/MFA, JWT/OAuth, RBAC (Casbin), CSRF/XSS/SSRF mitigation, rate limiting, audit logging; CodeQL, Bandit, Trivy, Gitleaks

Cloud, DevOps & Testing: AWS (ECS Fargate, Amplify, S3, SES, CloudWatch), GCP, Docker, Nginx, GitHub Actions, CI/CD, Playwright

ADDITIONAL

Programs: Citigroup Financial Investment Analysis Training Program, Shanghai (Jul 2023) — selective cohort; built Excel and Python models for company-performance analysis.

Conferences: Google Cloud Next 2026; Google I/O 2025; Microsoft AI Day 2024.